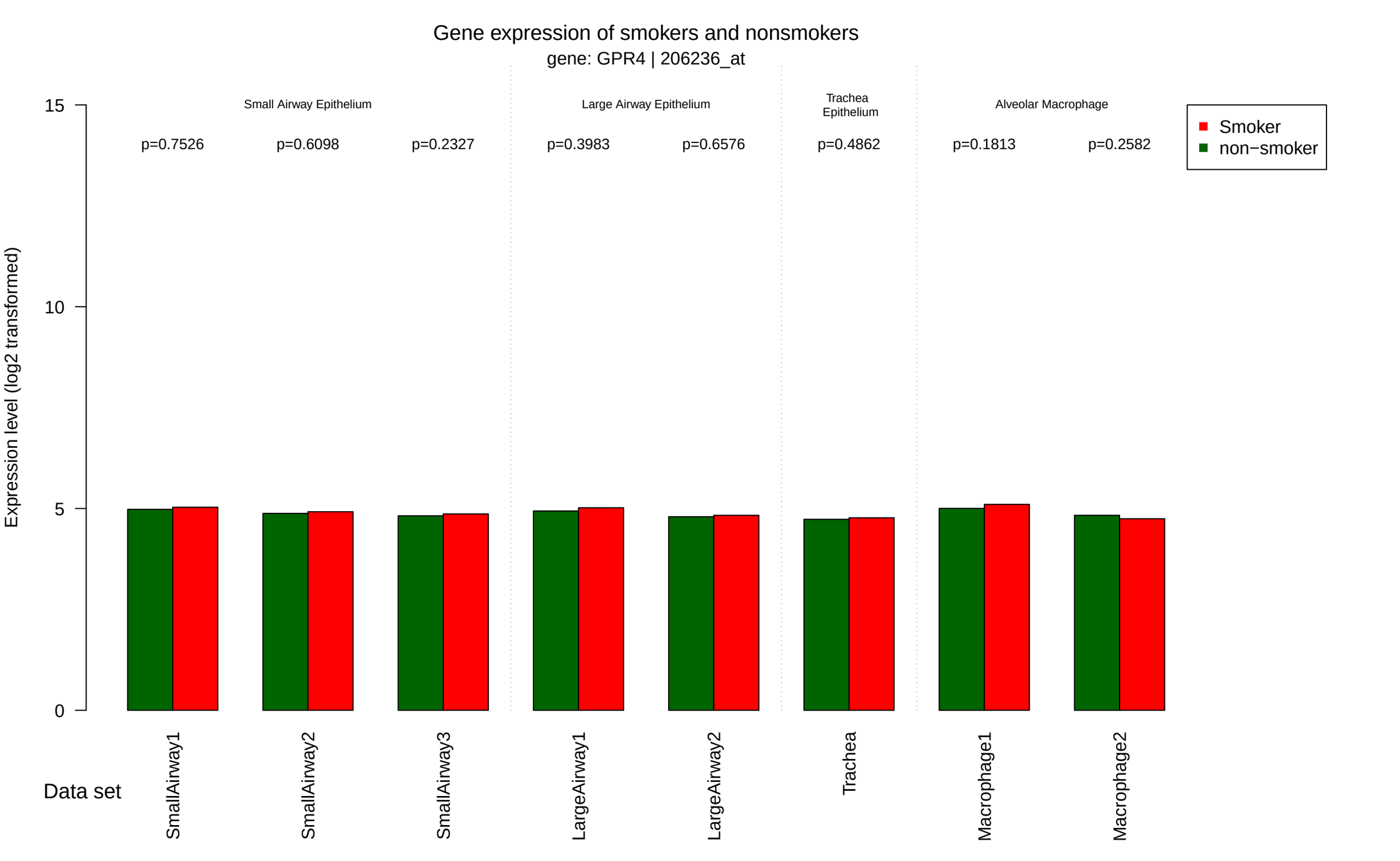
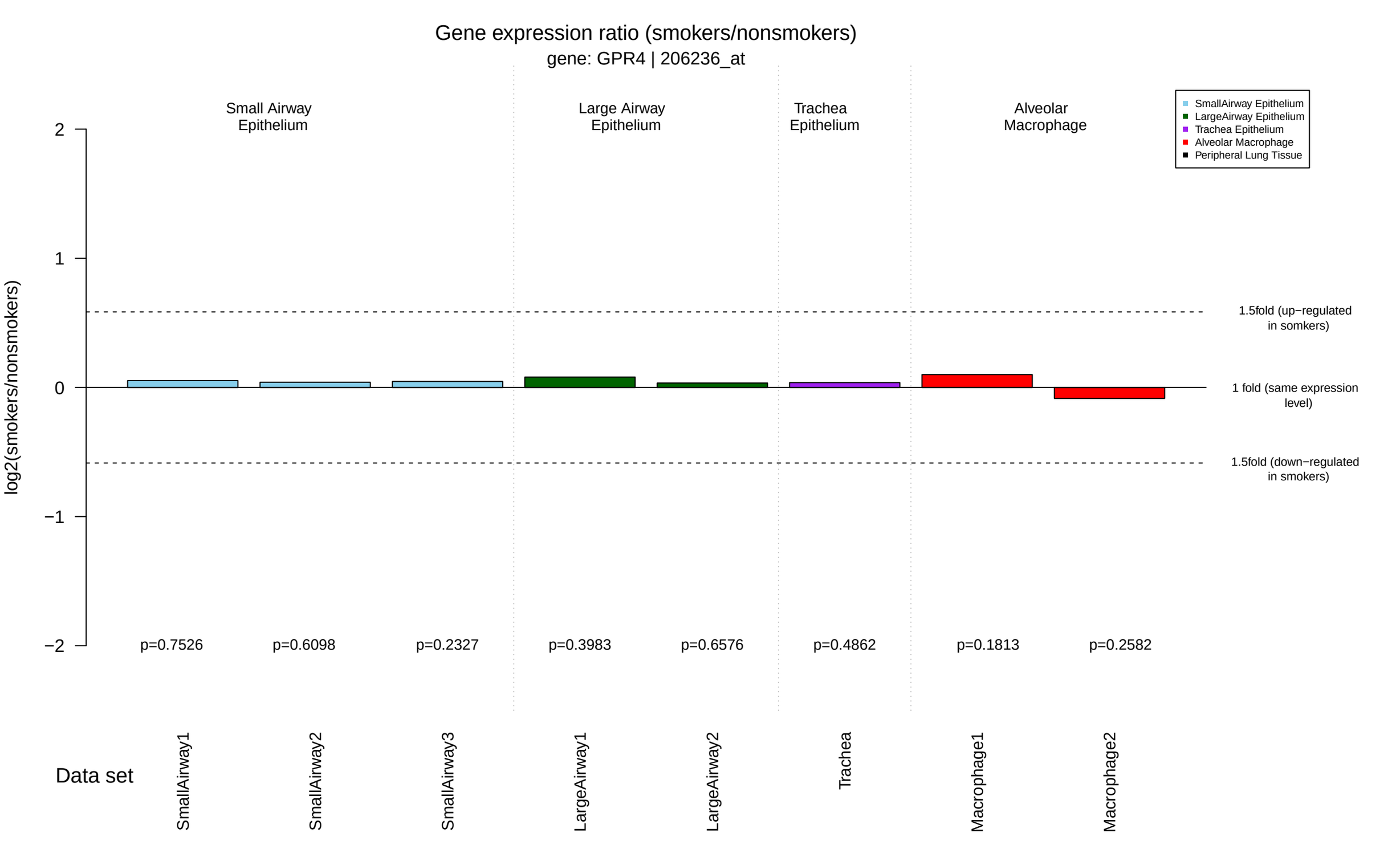
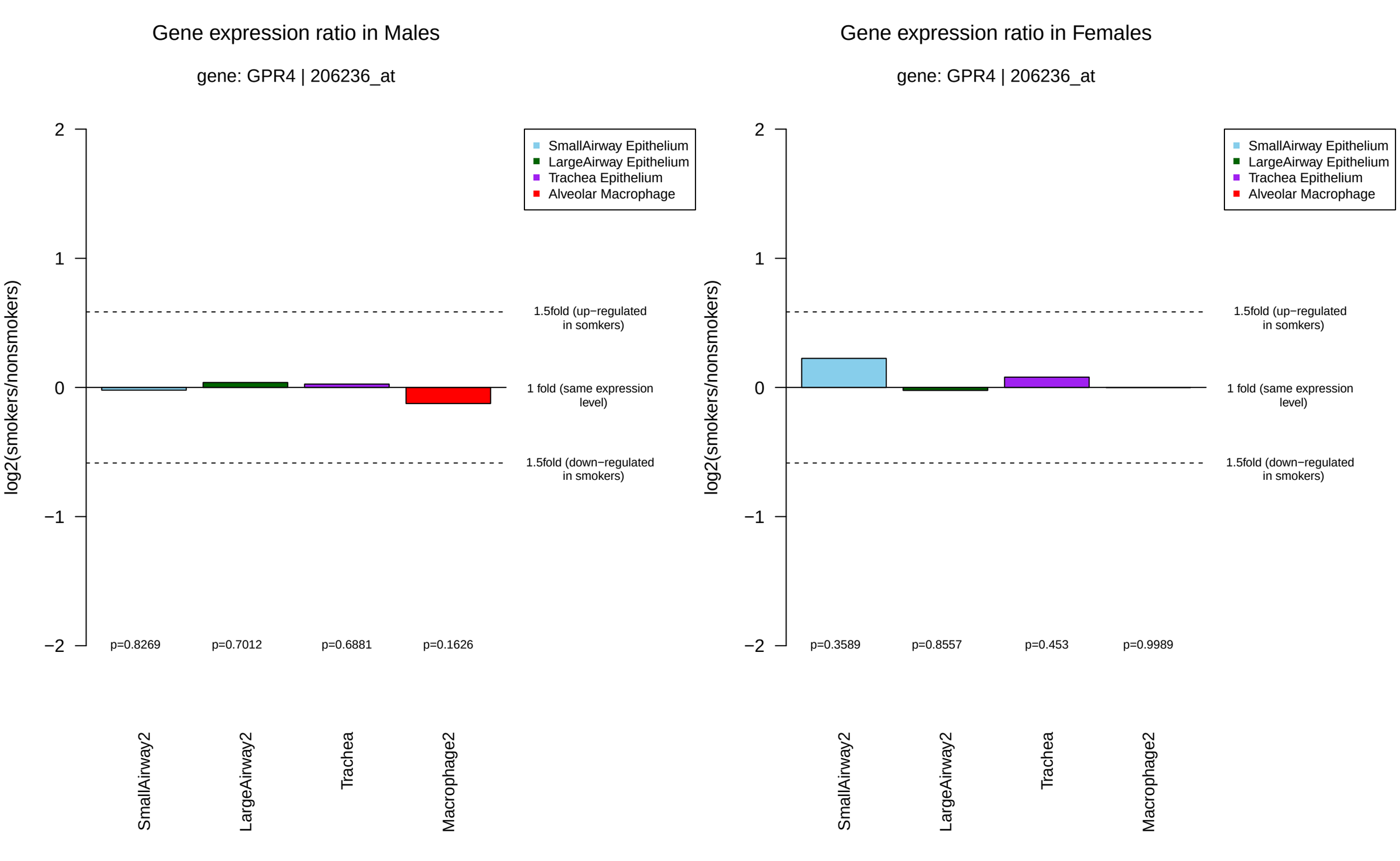
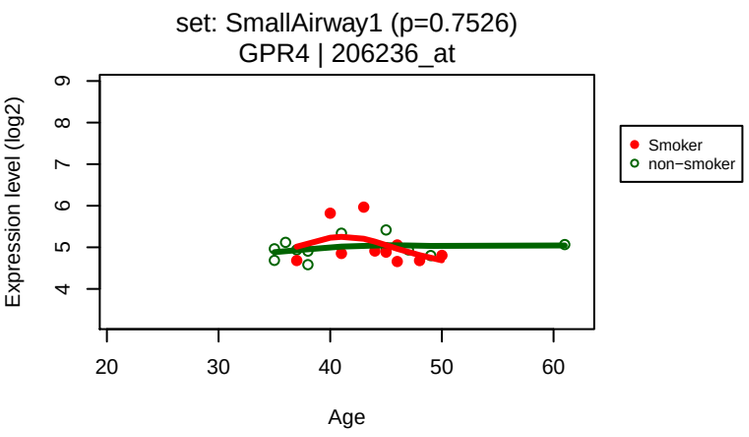
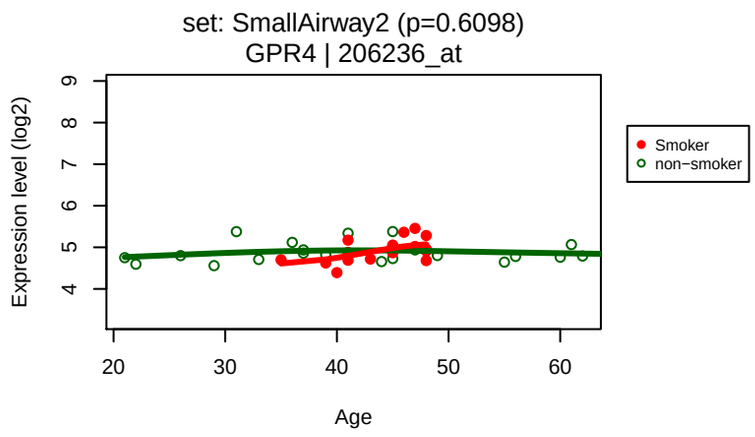
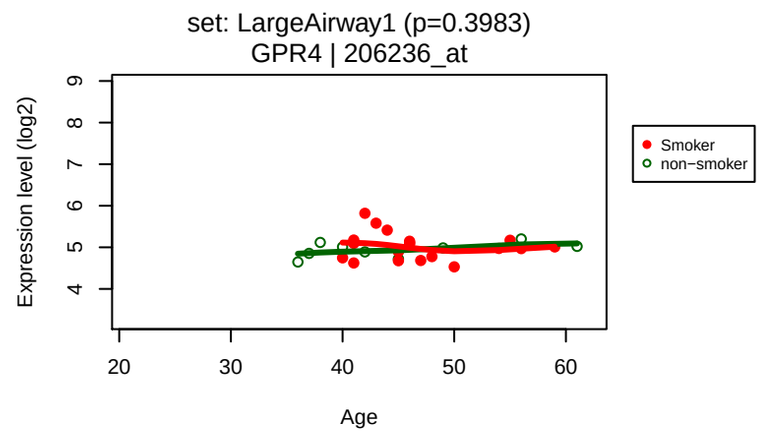

Gene expression of smokers and nonsmokers
gene: GPR4 | 206236_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.7526
p=0.6098
p=0.2327
p=0.3983
p=0.6576
p=0.4862
p=0.1813
p=0.2582
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

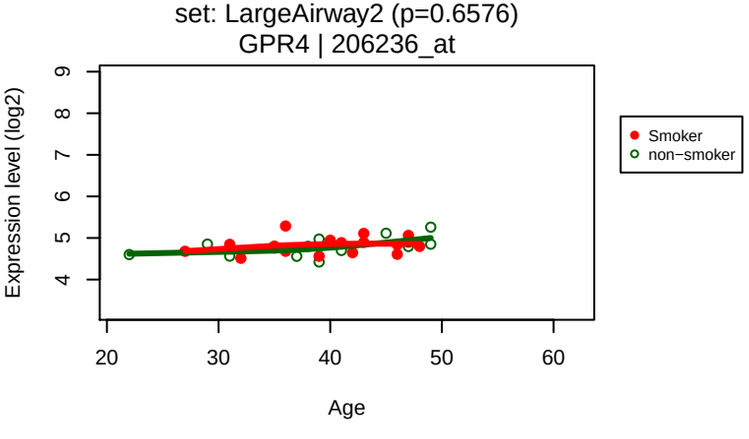

Gene expression ratio (smokers/nonsmokers)
gene: GPR4 | 206236_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7526
p=0.6098
p=0.2327
p=0.3983
p=0.6576
p=0.4862
p=0.1813
p=0.2582
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

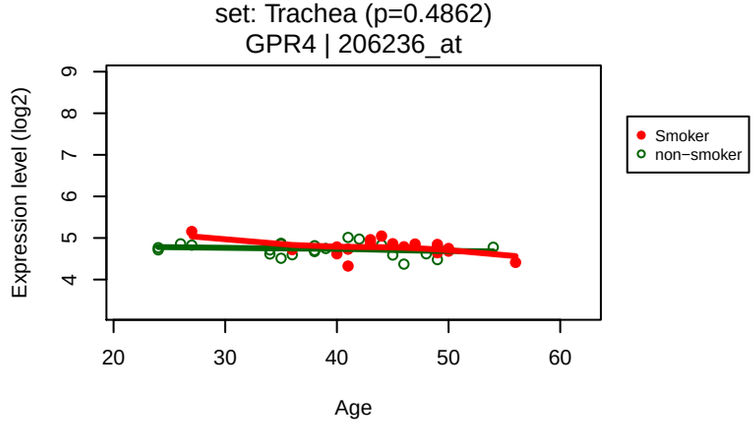

Gene expression ratio in Males
gene: GPR4 | 206236_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8269
p=0.7012
p=0.6881
p=0.1626
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GPR4 | 206236_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3589
p=0.8557
p=0.453
p=0.9989
SmallAirway2
LargeAirway2
Trachea
Macrophage2

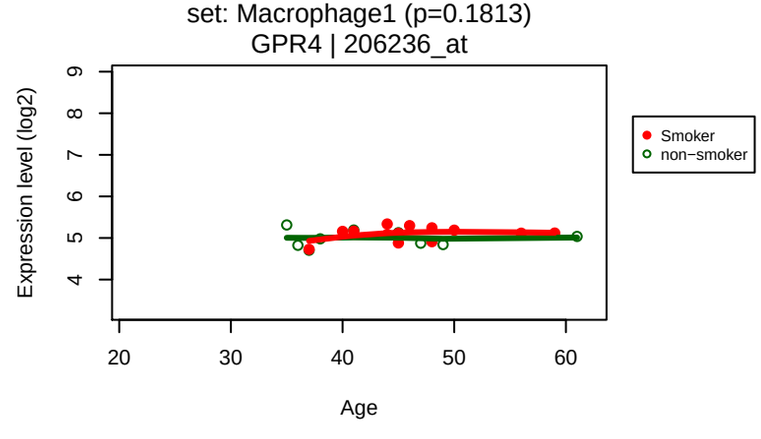

set: SmallAirway1 (p=0.7526)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker

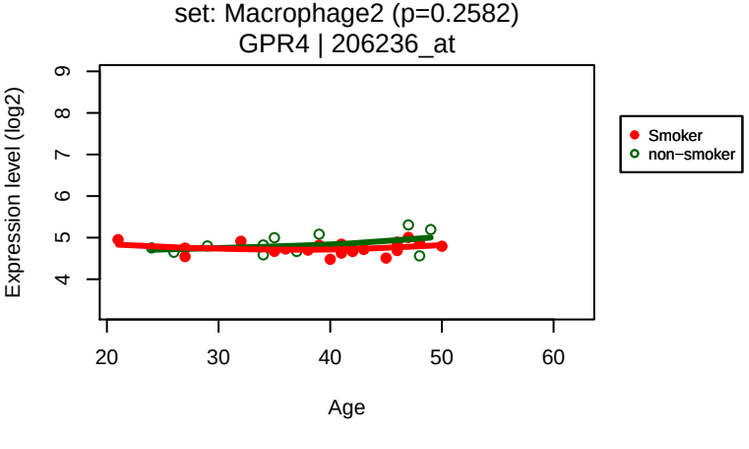

set: SmallAirway2 (p=0.6098)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker

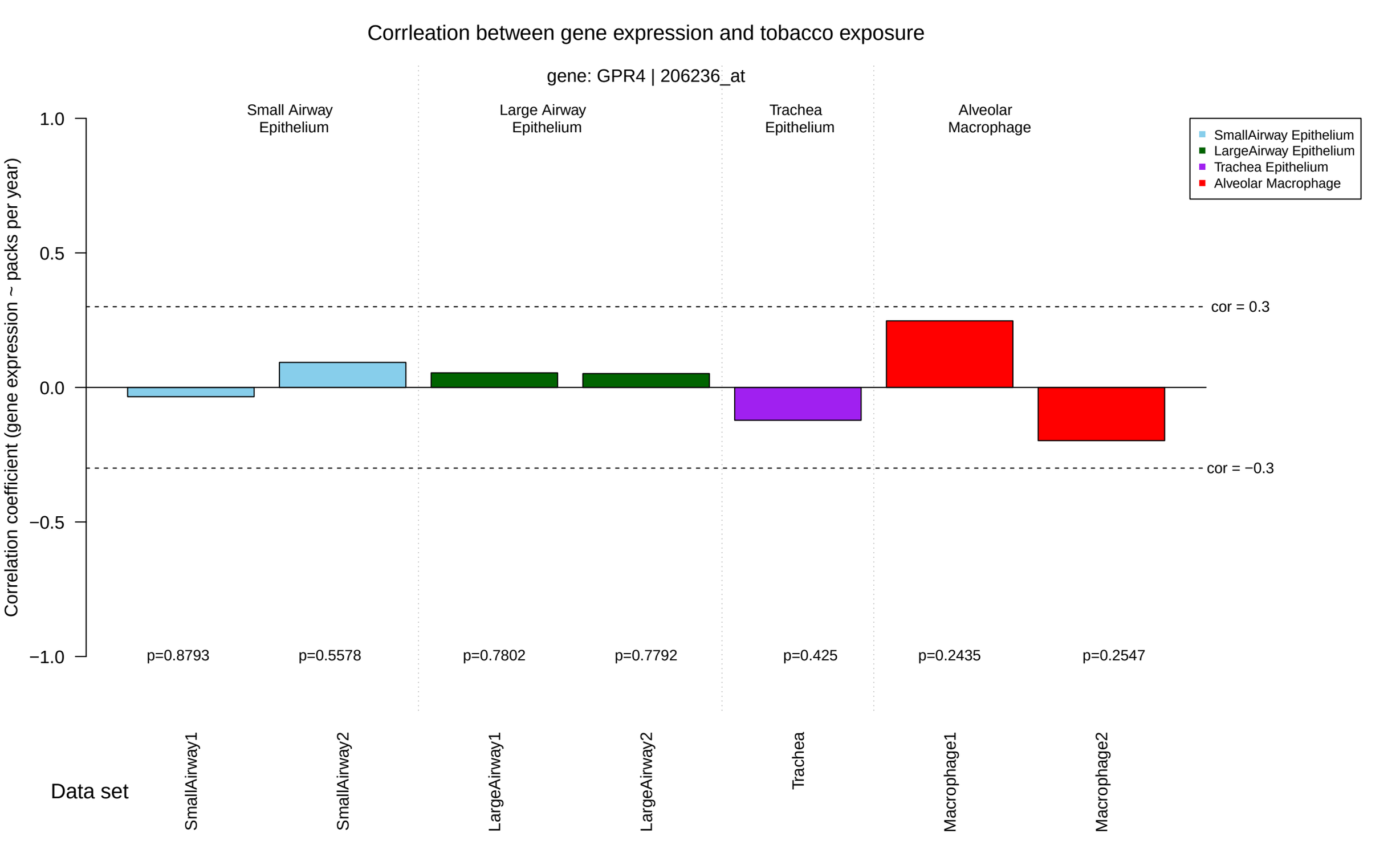

set: LargeAirway1 (p=0.3983)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6576)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4862)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1813)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2582)
GPR4 | 206236_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GPR4 | 206236_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8793
p=0.5578
p=0.7802
p=0.7792
p=0.425
p=0.2435
p=0.2547
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2